

Computational models of sequential decision-making: RNN studies

This study investigates how recurrent neural networks (RNNs) model sequential economic decision-making and whether incorporating spatial action mapping alters their internal dynamics. Building on Gu (2025), which showed that identical inputs can yield stable or rotational population patterns depending on output demands, the project extends the model by adding spatial location as an explicit input dimension. Using RNN, decoding analyses, and population-level dimensionality reduction, the study examines changes in choice encoding and latent circuit structure. By aligning model inputs more closely with real task conditions, this research aims to clarify how output contingencies and spatial mappings working memory and decision dynamics.

I. Research Question and Significance

Understanding how the brain makes decisions is important because decision-making shapes everyday behavior, from simple choices to complex economic judgments. To uncover the underlying mechanisms, researchers often use computational models. RNNs are especially useful tools because, like the brain, they have recurrent (feedback) connections and process information over time. This makes them well suited for studying decisions that unfold sequentially.

In daily life, many economic decisions unfold sequentially rather than simultaneously. When options are presented one after another, the brain must remember earlier information before making a choice. Previous research shows that decisions can be organized in different ways, for example, based on option identity (juice A vs. juice B) or based on presentation order (first vs. second). However, it is still unclear whether these different representations rely on the same neural mechanism.

Gu (2025)'s study trained recurrent neural networks (RNNs) on two similar sequential tasks that differed only in how choices were reported. Although the inputs were the same, the networks developed different internal activity patterns depending on the task. Identity-based choices led to stable activity states, while order-based choices showed rotating activity patterns. This suggests that task demands can shape how decisions are processed.

However, in real monkey experiments, choices are expressed through eye movements to the left or right side of the screen. This spatial action mapping may influence how value representations are transformed into motor decisions. The previous RNN model does not incorporate this spatial action information into the input structure. On account of this, this study aims to investigate that if the spatial information in the original monkey task is added into the RNN model, how will the model training result and high-dimensional analysis change.

This study is significant because computational models are constrained by the information we provide during training. If important task variables such as spatial action mapping are omitted, the model may differ from biological reality. By aligning the model's inputs more closely with the actual experimental conditions, we increase the likelihood that the model's internal dynamics reflect the true neural mechanisms.

II. Project Design and Feasibility

2.1 Project Methods

1. Use PsychRNN as the framework to define the decision-making process based on vanilla RNN (may be changed to structure of LSTM or GRU) which includes the variability of input and the expected form of output. ReLU is chosen as the activation function.
2. Use Linear Discriminant Analysis and RepeatedStratifiedKFold to decode the accuracy of the model's representation of information at different time points.
3. Quantify the choice preference using Receiver Operating Characteristic and Area Under the Curve for single neuron tuning analysis.
4. Visualize neural manifolds using Principal Component Analysis (TDR as a backup) and recording Rotation Index for population dynamics analysis.

5. Calculate the angle between the encoding vectors of different attributes and evaluate their degree of orthogonality.
6. Infer the low dimension connectivity structure from RNN activity.
7. Investigated the relationship between the encoding of asynchronously presented stimuli using Bivariate Linear Regression in both perceptual and economic decision-making.
8. Verify the trained RNN by comparing with lab data.
9. Examined the consistency of correlation patterns.

2.2 Timeline plan

Week	Stage	Specific Progress Goals
Week 1-2	Environment setup and modeling	Review literature in depth; Write code for generating multi-attribute stimuli; Build a basic RNN framework and initially run the juice/order task from the original paper.
Week 3	Attribute extension	Introduce spatial location input; Adjust network hyperparameters to ensure the model can extract "order" or "identity" reference frames under spatially distributed stimuli.
Week 4	Single-unit activity analysis	Analyze single-unit activity of different stages in RNN trained for each task to compare the new pattern of tuning with the original one.
Week 5-6	Dynamics in population activity	Utilize fixed-point analysis to examine the stability of the network during the delay period; Observe the mixed morphology of attractor and rotational dynamics.
Week 7-8	Latent circuit inference	Derive reduced circuit diagram; Analyze the consistency of choice representations between the RNN and its reduced circuit; Obtain trial-averaged activity.
Week 9-10	Data comparison	Evaluate the decoding accuracy across reference frames based on existing dataset; Explore the relationship between encoding and signal choice.
Week 11	Summary of Results	Complete the first draft of a technical paper, and summarize the working memory representation patterns in multi-attribute decision-making.

III. Background

Both Zhangcheng Li and Herui Lyu have taken Introduction to Neuroscience, Machine Learning, and Introduction to Computer and Data Science at school. At the same time, they have completed Introduction to Computational Neuroscience on Coursera and read books such as “Spikes, Decisions, and Actions” recommended by Professor Sukbin Lim.

In addition, Yiwen Xu has taken Behavior and Integrative Neuroscience, Cellular and Molecular Neuroscience, Computational Neuroscience, Data management and analysis, Data structure course. Meanwhile, Yiwen assisted in other computational neuroscience projects in a lab in New York.

IV. Group Partnership Plan

Our team brings together diverse perspectives: Yiwen Xu (Junior, Neuroscience and Data Science) is currently studying away in New York, while Zhangcheng Li (Sophomore, Neuroscience and Maths) and Herui Lyu (Sophomore, Computer Science) are based locally. Despite the distance, we maintain a highly collaborative environment through a WeChat group for real-time communication and twice-per-week Zoom meetings to synchronize our progress.

Based on our respective expertise, we have defined our roles as follows: Herui Lyu and Zhangcheng Li will focus on computational tasks, including model training and data analysis; Yiwen Xu will lead the theoretical research relevant to the project's scope. However, these assignments are not mutually exclusive. To ensure project coherence and integrity, we will maintain close communication and monitor each other's progress throughout the collaboration.

V. Feedback, Evaluation and Dissemination of Knowledge

Under the mentorship of Professor Sukbin Lim, the project will benefit from bi-weekly consultations to ensure consistent expert evaluation regarding experimental rigor, data interpretation, and theoretical implications. These regular meetings will serve as a critical feedback loop, guiding the study toward its culmination in a comprehensive final report. Furthermore, should the results yield substantive insights, we intend to disseminate our findings through conference presentations or submissions to peer-reviewed academic journals.

VI. Reference

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